

	
Iniziato	3 settembre 2025, 11:05
Stato	Completato
Terminato	3 settembre 2025, 12:20
Tempo impiegato	1 ora 15 min.
Valutazione	9,88 su un massimo di 18,00 (55%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12

Domanda 1

Risposta
errata

Punteggio
ottenuto 0,00
su 1,50

1.5 points (no penalty for a wrong answer)

You are given an encoder-only model. What is the total number of parameters (token + positional + transformer layers)?

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters and task-specific heads.

- Vocabulary size $V=128,000$
- Embedding/model dimension $d=768$
- Positional embeddings: absolute, learned, for 2048 positions
- Number of layers $L=36$
- Number of attention heads $H=12$
- FF-NN projection dimension: 3072
- Head dimension $d_k = d_v = 64$

Write the answer in the box below, **in millions of parameters** (round to the closest million).

For instance, if the answer is 123,456,789 parameters, write 123.

Risposta: 

La risposta corretta è : 98

Domanda 2

Completo

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for wrong answers)

Consider the following cosine similarity table. Which of the following threshold is more dependable in terms of TPR and FPR?

$T = 0.75, 0.85, 0.95$

	Exp1	Exp2	Exp3	Exp4
Act1	0,88	0,74	0,77	0,60
Act2	0,44	0,76	0,71	0,20
Act3	0,63	0,54	0,73	0,33
Act4	0,71	0,68	0,79	0,83

$TPR = TP / TP + FN;$

$FPR = FP / FP + TN;$

$T = 0.75$ --> 0.75 is the more dependable threshold

$TPR = 3 / 4$

$FPR = 2 / (2 + 10)$

$T = 0.85$

$TPR = 1 / 4$

$FPR = 0$

$T = 0.95$

$TPR = 0 / 4$

$FPR = 0$

Commento:

ok

Domanda 3

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for a wrong answer)

The main goal of *model distillation* is:

- ☐ a. To increase the model's tokenization speed by optimizing the vocabulary size
- ☐ b. To enhance the model's ability to generate longer sequences without truncation
- ☐ c. To focus on improving the model's understanding of rare or unseen tokens
- ☐ d. None of the others is the main goal of model distillation
- ☐ e. To train language models to independently learn new languages from scratch
- ☐ f. To fine-tune language models to follow explicit instructions
- ☒ g. To reduce the size of the model without sacrificing performance ✓

Your answer is correct.

La risposta corretta è:

To reduce the size of the model without sacrificing performance

Domanda 4

Parzialmente
corretta

Punteggio
ottenuto 0,50
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following is/are **not** a technique to evaluate the quality of test cases generated with LLMs?

- ☐ a. Mutation score
- ☐ b. Precision and recall
- ☐ c. Coverage
- ☒ d. Functional Correctness ✓
- ☐ e. Execution Success Rate

Risposta parzialmente esatta.

Hai selezionato correttamente 1.

Le risposte corrette sono:

Functional Correctness,

Precision and recall

Domanda 5

Completo

Punteggio
ottenuto 2,50
su 3,00

3 points (no penalty for a wrong answer)

The scenario-based requirements engineering (SRE) is a technique where requirements are constructed in three steps:

- identification of **actors**
- description of **scenarios**
- extraction of **functional requirements**

Here, actors are the entities (people, systems, or organizations) that interact with the system, while scenarios are narrative descriptions of how actors use the system to achieve specific goals.

Consider the case where you receive natural language requirements in a string (called REQS). Describe an architecture of LLAMA agents to perform the steps of the SRE technique, and the prompt engineering techniques employed in each step.

For each agent, write the prompt including System, context, possible inputs from previous prompts, and expected output.

Notice: it is not necessary to write the python calls to the transformers or code for output parsing.

Architecture description:

Agent 1 extract entities that are involved in the Input REQS using Router prompting.

Agent 2 Given the actors extracted by Agent 1, the input REQS and Few examples generate a narrative description on what the actor does in the given situation

Agent 3 Given the scenarios generate by Agent 2 extract the functional requirements, Few-show prompt with examples of scenarios + extracted features

Prompt engineering techniques adopted:

Router prompting for Agent 1

Few-Show prompting for Agent 2 and 3

Prompts:

```
prompt1 = ""
```

you are a powerfull AI that routes Software engineering tasks your goal is to understand which are the entities involved in the given instruction:

If instruction is about "people", output:

"people"

If intruction is about "systems", output:

"system"

If instruction is about "organizations", output:

"organizations"

If instruction is not about any of above, output:

"none of above"

Output:

```
""
```

```
prompt2 = "" you are an expert on Software engineering requirements specifically for generating scenarios
given actors. your goal is to generate scenarios in natural language.
```

```
output <add output formats/costraints/etc>
```

```
***Example 1***
```

```
input_prompt="blalblalbla + <actors extracted> act1, act2, act2</actors extracted>"
```

```
generated_scenarios=["this is scenario for actor1", "another scenario for act1", "scenario for act2"....]
```

```
...
```

```
*** Example X *** (X is decided accordi to the context windows length, It can be just 2 or 10)
```

```
input_prompt="blalblalbla + <actors extracted> act1, act2, act2</actors extracted>"
```

```
generated_scenarios=["this is scenario for actor1", "another scenario for act1", "scenario for act2"....]
```

Now perform the task given the following:

```
<input>Input REQS</input>
```

```
<actors extracted> Response from Agen1</actors extracted>
```

Output:

```
""
```

```
prompt3 = you are an expert on Software engineering requirements tasks. your goal is to extract the
functional requirements given the input REQS, the scenarios and the extracted actors.
```

```
output <add output formats/costraints/etc>
```

```
***Example 1***
```

```
input_prompt="<input REQS> + <actors extracted> act1, act2, act2</actors extracted> + <scenarios>Here
generated scenarios</scenarions>"
```

```
extracted_funcReq=[func_req_1, func_req_2 ...]
```

```
...
```

```
*** Example X *** (X is decided accordi to the context windows length, It can be just 2 or 10)
```

```
input_prompt="<input REQS> + <actors extracted> act1, act2, act2</actors extracted> + <scenarios>Here
generated scenarios</scenarions>"
```

```
extracted_funcReq=[func_req_1, func_req_2 ...]
```

Now perform the task given the following:

```
<input>Input REQS</input>
```

```
<actors extracted> Response from Agen1</actors extracted>
```

```
<scenarios> Response from Agent2>/scenarios>
```

output:

```
""
```

Commento:

Output of prompt 1 is not precise, since it only outputs "actor" or "system" and not the actual name of the actor or system that is necessary to formulate the requirements

Domanda 6

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

You are given the following sequence of characters.

b c a c b c a b b b b c

Initially, each character is a separate token.

Apply **Byte-Pair Encoding** to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

1) if the merged tokens are aa and b, write: **aa,b**

2) if the merged tokens are x and y, write: **x,y**

1st merge: b,b



Risposta corretta

La risposta corretta è : b,b

Punteggio ottenuto 1,00 su 1,00

2nd merge: b,c



Risposta corretta

La risposta corretta è : b,c

Punteggio ottenuto 1,00 su 1,00

3rd merge: bc,a



Risposta corretta

La risposta corretta è : bc,a

Punteggio ottenuto 1,00 su 1,00

Domanda 7

Risposta
errata

Punteggio
ottenuto -0,38
su 1,50

1.5 points (-25% penalty for a wrong answer)

Which of the following best describes how BERTScore evaluates text similarity?

- ☐ a. It applies a bag-of-words model to compare token frequency distributions between reference and candidate texts
- ☐ b. It uses a fine-tuned BERT model to classify whether the reference and candidate texts are semantically related or not
- ☒ c. It uses cosine similarity on static word embeddings trained with Word2Vec or GloVe ✖
- ☐ d. It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores
- ☐ e. It measures sequence similarity using n-gram co-occurrence weighted by IDF scores
- ☐ f. It predicts a similarity score by treating the evaluation task as a masked language modeling problem, where BERT predicts the score as the next token
- ☐ g. It assigns similarity scores based on syntactic parse trees rather than token embeddings

Your answer is incorrect.

La risposta corretta è:

It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores

Domanda 8

Parzialmente
corretta

Punteggio
ottenuto 0,75
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following are benefits of a Retrieval-Augmented-Generation approach to prompting?

- ☐ a. It is possible to adopt self-critique mechanisms to refine the outputs of the agent.
- ☐ b. It is possible to decompose a task in several smaller steps.
- ☒ c. It is possible to prevent the obsolescence of answers by using up-to-date documents. ✓
- ☒ d. It is possible to reduce the amount of false information provided to the final user. ✓
- ☒ e. It is possible to reduce the amount of hallucinations by providing a fixed structure for the agent's answers. ✗

Risposta parzialmente esatta.

Hai selezionato troppe opzioni.

Le risposte corrette sono:

It is possible to reduce the amount of false information provided to the final user.,

It is possible to prevent the obsolescence of answers by using up-to-date documents.

Domanda 9

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

What is "priming" in the context of prompt engineering?

- ☐ a. Pre-training the model with additional data
- ☐ b. Adding noise to the input to test model robustness
- ☐ c. Resetting the model before every new prompt
- ☒ d. Providing initial input to guide the model's response ✓
- ☐ e. Using reinforcement learning to fine-tune the model

Risposta corretta.

La risposta corretta è:

Providing initial input to guide the model's response

Domanda 10

Risposta
errata

Punteggio
ottenuto -0,25
su 1,00

1 point (-25% penalty for each wrong answer)

Adapter layers are introduced to reduce the number of tunable parameters. They consist in:

- ☐ a. Replacing the feedforward layers in Transformers with low-rank matrix decompositions to reduce memory usage
- ☐ b. None of the others is a valid description of adapters
- ☐ c. Additional layers that replace the attention mechanism in the Transformer architecture to simplify computation
- ☐ d. Small, trainable layers inserted into a pre-trained model, which are fine-tuned together with the rest of the original model
- ☐ e. A method for splitting a large model into smaller sub-models that are trained independently and later combined
- ☐ f. A technique to merge multiple pre-trained models into a single model with fewer parameters

- ☒ g. A task-specific head added to the model, which is fine-tuned while the rest of the model is kept frozen



Your answer is incorrect.

La risposta corretta è:

None of the others is a valid description of adapters

Commento:

**Domanda
11**

Parzialmente
corretta

Punteggio
ottenuto 0,25
su 2,50

2.5 points (no penalty for a wrong answer)

The attention for matrices Q , K , V (queries, keys, values) is computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where d_k is the dimensionality of the keys.

The following are the input tokens that reach an attention layer:

$$x = \begin{bmatrix} 2 & 3 & 3 \\ 0 & 3 & 1 \end{bmatrix}$$

The following are the W_q , W_k , W_v matrices for the attention layer:

$$W_q = \begin{bmatrix} -2 & -1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$W_k = \begin{bmatrix} 0 & -2 \\ -1 & 0 \\ -1 & 1 \end{bmatrix}$$

$$W_v = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Compute the output of the attention operation for the first token.

Write the answer in the following boxes (one dimension for each box).

Report at least 4 significant figures.

Dimension 1:



Risposta non data
La risposta corretta è : 3
Punteggio ottenuto 0,00 su 1,00

Dimension 2:



Risposta non data
La risposta corretta è : -3
Punteggio ottenuto 0,00 su 1,00

Dimension 3:



Risposta non data
La risposta corretta è : -5
Punteggio ottenuto 0,00 su 1,00

Commento:

**Domanda
12**

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

What is the purpose of reflection in an agentic system?

- ☒ a. To evaluate past decisions and improve future performance ✓
- ☐ b. To store knowledge permanently in short-term memory
- ☐ c. To eliminate the need for structured workflows
- ☐ d. To generate human-like responses using predefined scripts
- ☐ e. To execute tasks without modifying strategies

Risposta corretta.

La risposta corretta è:

To evaluate past decisions and improve future performance